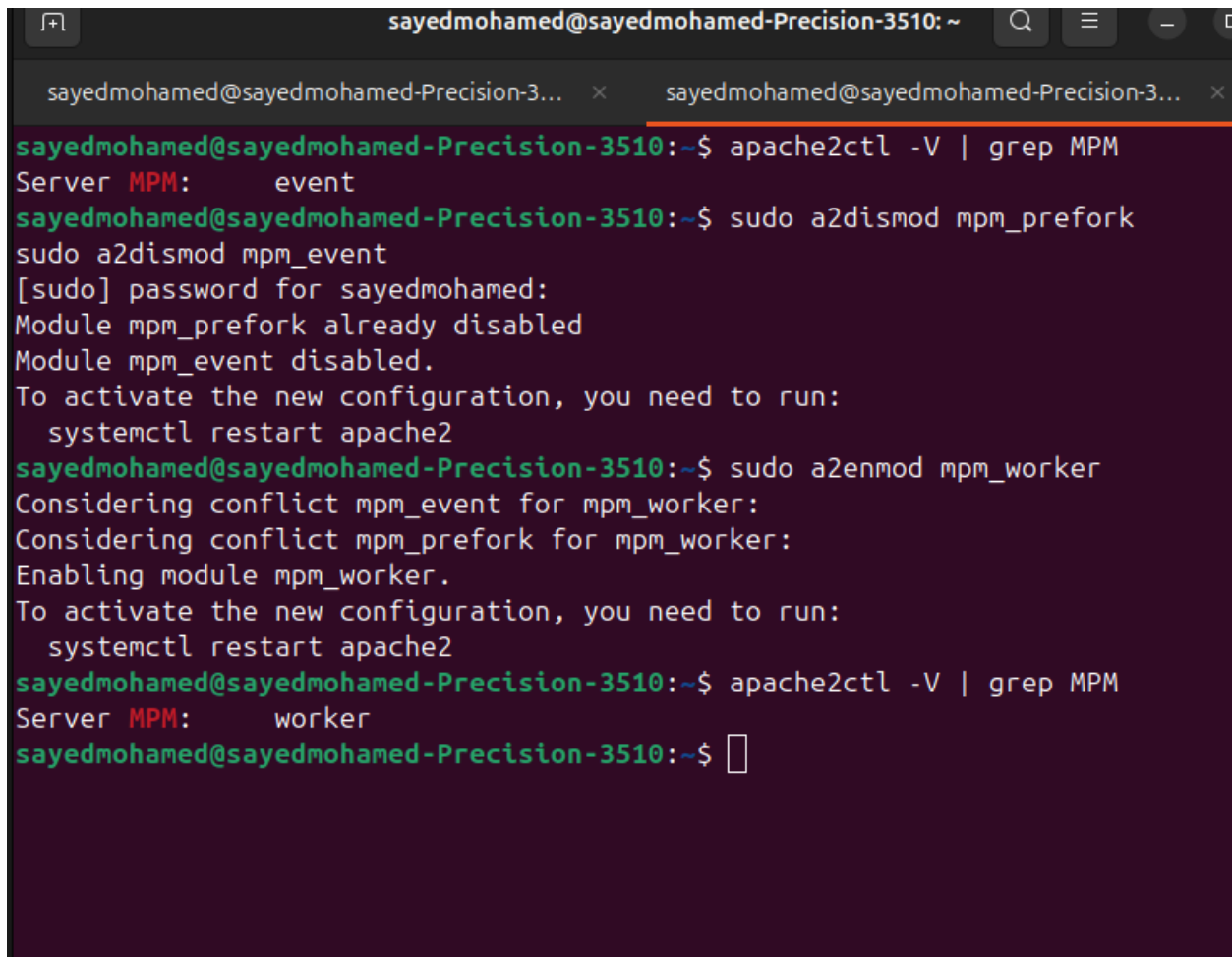


Elsayed mohamed elsayed farg

Apache Lab2

1-Configure apache to use Worker MPM



```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3510:~$ apache2ctl -V | grep MPM  
Server MPM:      event  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo a2dismod mpm_prefork  
sudo a2dismod mpm_event  
[sudo] password for sayedmohamed:  
Module mpm_prefork already disabled  
Module mpm_event disabled.  
To activate the new configuration, you need to run:  
    systemctl restart apache2  
sayedmohamed@sayedmohamed-Precision-3510:~$ sudo a2enmod mpm_worker  
Considering conflict mpm_event for mpm_worker:  
Considering conflict mpm_prefork for mpm_worker:  
Enabling module mpm_worker.  
To activate the new configuration, you need to run:  
    systemctl restart apache2  
sayedmohamed@sayedmohamed-Precision-3510:~$ apache2ctl -V | grep MPM  
Server MPM:      worker  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

2-Check the status of rewrite_mod and enable it if it is enabled.

```
sayedmohamed@sayedmohamed-Precision-3510: ~  
sayedmohamed@sayedmohamed-Precision-3... x sayedmohamed@sayedmohamed-Precision-3... x  
sayedmohamed@sayedmohamed-Precision-3510:~$ apache2ctl -M | grep rewrite  
AH00558: apache2: Could not reliably determine the server's fully qualified  
in name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress  
is message  
rewrite_module (shared)  
sayedmohamed@sayedmohamed-Precision-3510:~$
```

3-Write RewriteCond and RewriteRules for:

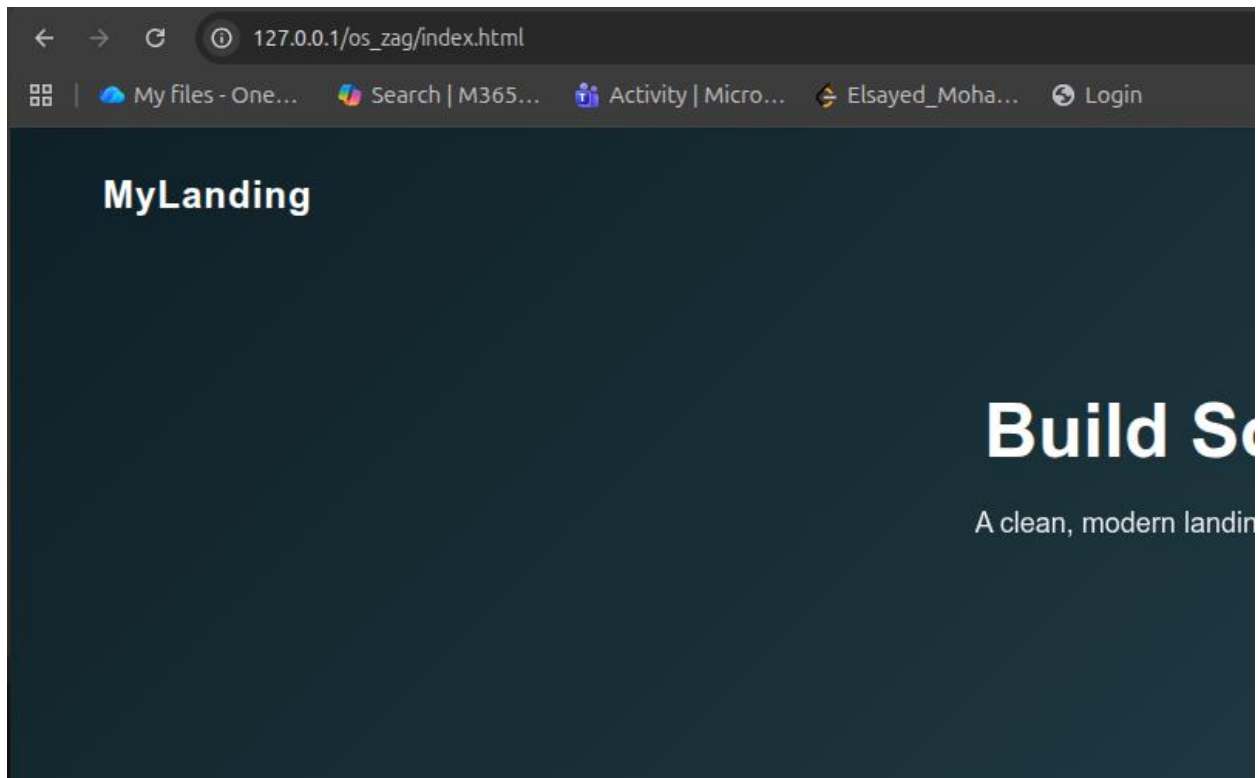
- a. Deny access to <http://localhost/page?queryString> if
queryString contains the string forbidden.

The image shows a terminal window at the top and a web browser window at the bottom. The terminal window is running GNU nano 7.2 and editing the file /etc/apache2/sites-available/000-default.conf. The content of the file is as follows:

```
##### Lab 2 #####  
#Question 3-a  
    RewriteCond %{QUERY_STRING} forbidden  
    RewriteRule ^/os_zag/index\.html$ - [F]  
  
</VirtualHost>
```

The web browser window shows the URL 127.0.0.1/os_zag/index.html?test=forbidden. The page displays a "Forbidden" error message: "You don't have permission to access this resource." Below the message, it says "Apache/2.4.58 (Ubuntu) Server at 127.0.0.1 Port 80".

b. Remove the Query String

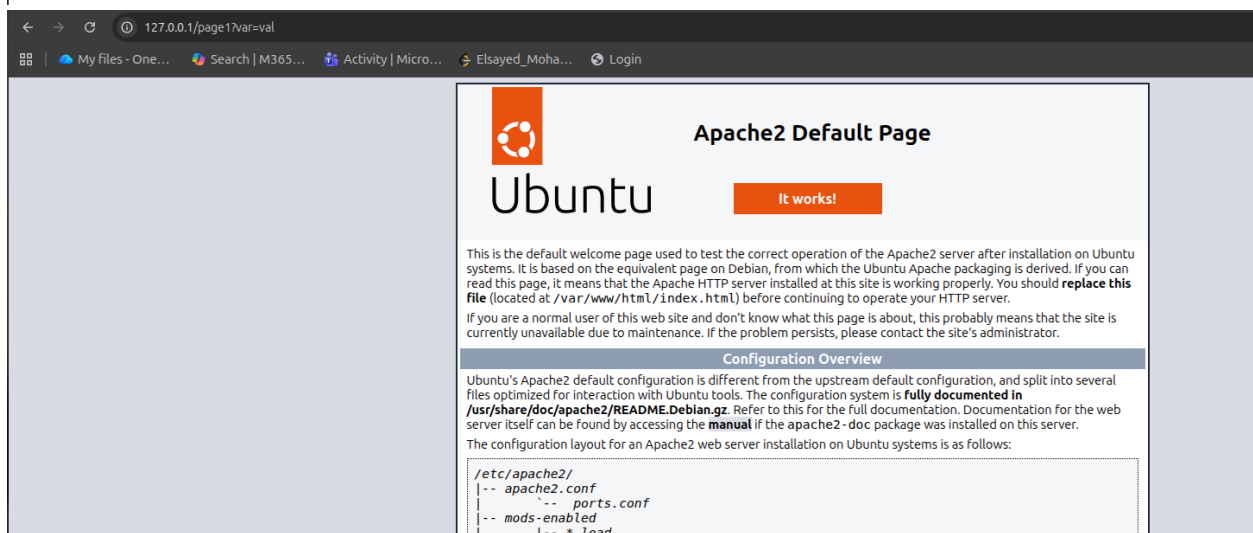
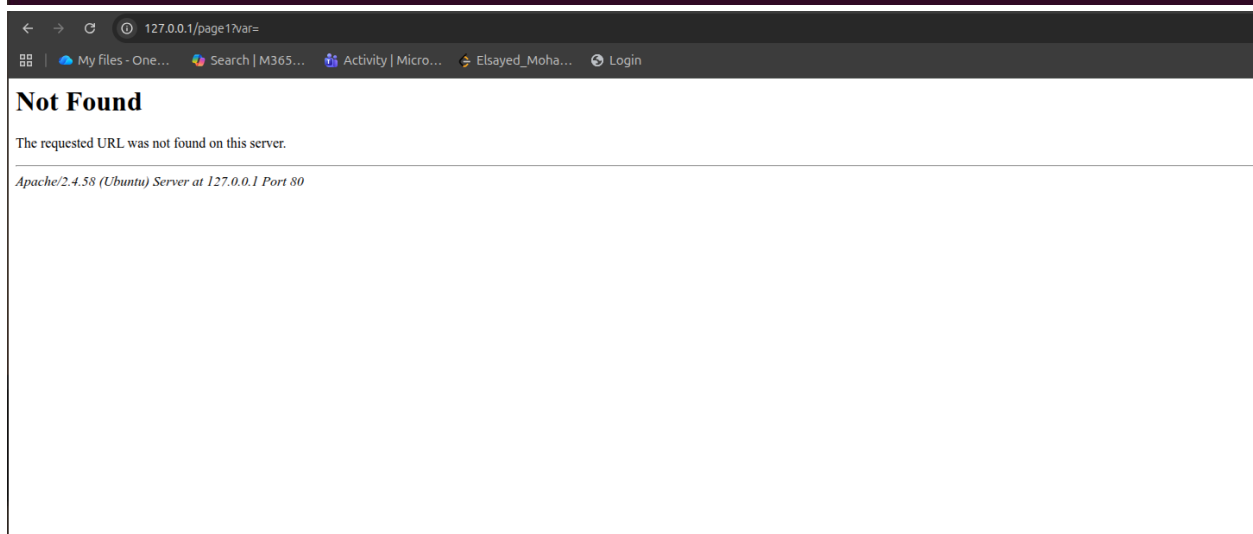


c. Rewrite URLs like `http://localhost/page1?var=val` to `http://localhost/page2?var=val` but do not rewrite if `val` isn't present.

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *

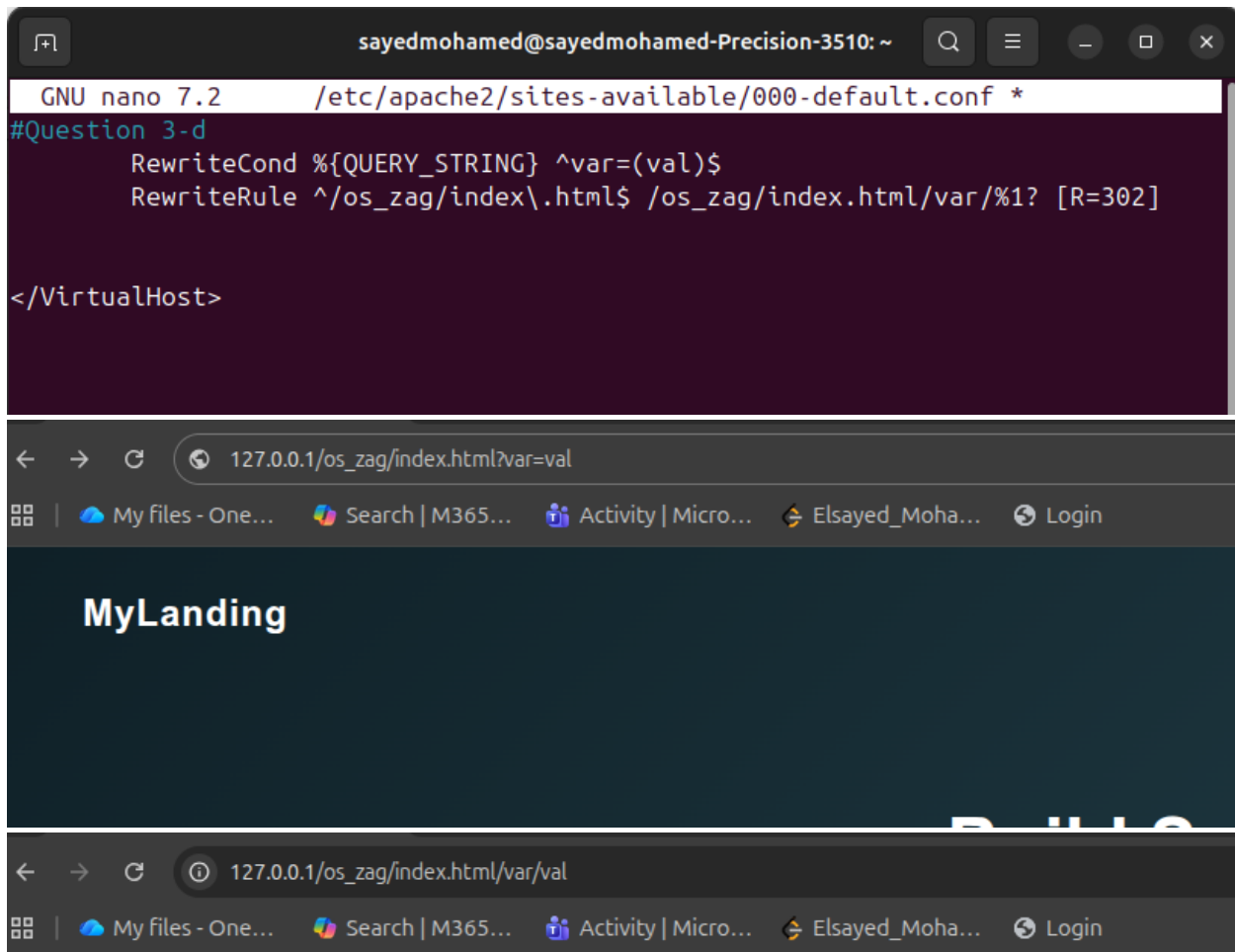
#Question 3-c
RewriteCond %{QUERY_STRING} ^var=val$
RewriteRule ^/page1$ /index.html

</VirtualHost>
```



d. Take a URL of the form `http://localhost/path?var=val` and transform it into

<http://localhost/path/var/val>.



Not Found

The requested URL was not found on this server.

Apache/2.4.58 (Ubuntu) Server at 127.0.0.1 Port 80

e. Map `http://localhost/example/one/two` to

<http://localhost/something.cgi?arg=one&other=two>

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *
#Question 3-e
RewriteRule ^/os_zag/index\.html/([^\./]+)/([^\./]+)$ /os_zag/index.html?arg=$1&other=$2

</VirtualHost>
```